

A Packing of 14 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 14 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.740953. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.726912, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 14 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.740953 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 14$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.726912, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 14/13 = 2.741259$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N14.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
685238382887611089669471900910299520282767613452431223901281223592945030
644855113933232948541235460657040991417974146093492652169678275086581578
749484814638536220559301604729068250683188267637
```

denominator:

```
2499999999999304055398893852790118461564929658120148053855400089117594211
028206100465806537776408499418128712813248382232371871047433096983257093
646426701112415029542365932476080199510624239163
```

B Coordinates

14 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N14.txt`.

```
153407354386 -89769584616 -499436763050 202312477715868671385699 71160878536 200460490757 267567561349-421333905376 31905400241 -715460647605 -415913258453
127703057516 153326138363 23898131067 -196010923356-64885963028 -241545389242 -777132736219 31708208919271087332 168464782963 -294519917007 187820565508
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